

Physical Chemistry (I)

Lecture 11

Maxwell-Boltzmann Statistics



In the Last Class

- Helmholtz and Gibbs energies
 - Definitions
 - Meanings
- Thermodynamics potentials
 - Differential forms
 - Properties of exact differentials
- Thermochemistry of Gibbs energy



Last class

Helmholtz and Gibbs Energies

$$\text{Helmholtz energy } A = U - TS$$

$$\text{Gibbs energy } G = H - TS$$

Criteria of spontaneity:

$$dA_{T,V} \leq 0$$

(At constant T and V , $dA \propto -dS_{\text{total}}$)

$$dG_{T,p} \leq 0$$

(At constant T and p , $dG \propto -dS_{\text{total}}$)



Last class

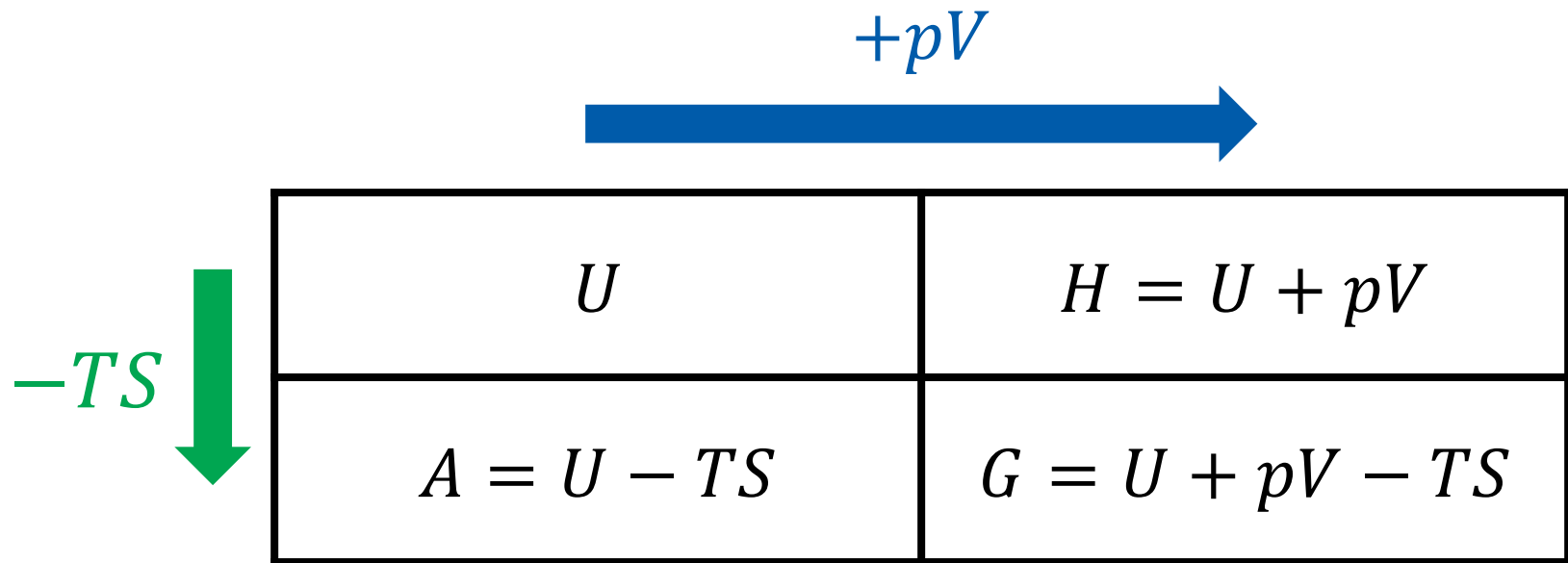
So, What Do They Mean?

- Helmholtz energy = maximum work
- Gibbs energy = maximum non-expansion work



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Thermodynamic Potentials



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Natural Independent Variables

- Internal energy $U = U(S, V)$

$$dU = TdS - pdV$$

- Enthalpy $H = U + pV = H(S, p)$

$$dH = TdS + Vdp$$

- Helmholtz energy $A = U - TS = A(T, V)$

$$dA = -SdT - pdV$$

- Gibbs energy $G = U + pV - TS = G(T, p)$

$$dG = -SdT + Vdp$$



Last class

Gibbs-Helmholtz Equation

$$\left(\frac{\partial(G/T)}{\partial T} \right)_p = -\frac{H}{T^2}$$

At constant T ,

$$\left(\frac{\partial(\Delta G/T)}{\partial T} \right)_p = -\frac{\Delta H}{T^2}$$



Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Internal energy $dU = TdS - pdV$

$$\left(\frac{\partial T}{\partial V}\right)_S = -\left(\frac{\partial p}{\partial S}\right)_V$$

- Enthalpy $dH = TdS + Vdp$

$$\left(\frac{\partial T}{\partial p}\right)_S = \left(\frac{\partial V}{\partial S}\right)_p$$

Maxwell Relations

For an exact differential $df = gdx + hdy$,

$$\left(\frac{\partial g}{\partial y}\right)_x = \left(\frac{\partial h}{\partial x}\right)_y$$

- Helmholtz energy $dA = -SdT - pdV$

$$\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$$

- Gibbs energy $dG = -SdT + Vdp$

$$\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$$

Last class

Thermodynamic Eq. of State

$$p = T \left(\frac{\partial p}{\partial T} \right)_V - \pi_T$$

Thermochemistry of G

$$G = H - TS$$

At constant T ,

$$\Delta G = \Delta H - T\Delta S$$

We can define the **standard Gibbs energy of reaction**:

$$\Delta_r G^\ominus = \Delta_r H^\ominus - T\Delta_r S^\ominus$$

and the **standard Gibbs energy of formation**:

$$\Delta_f G^\ominus = \Delta_f H^\ominus - T\Delta_f S^\ominus$$

Last class

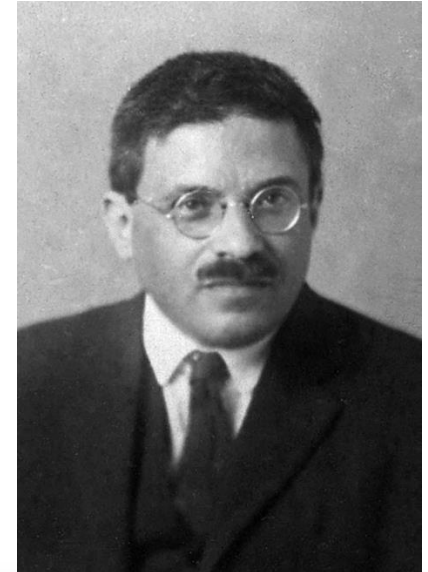
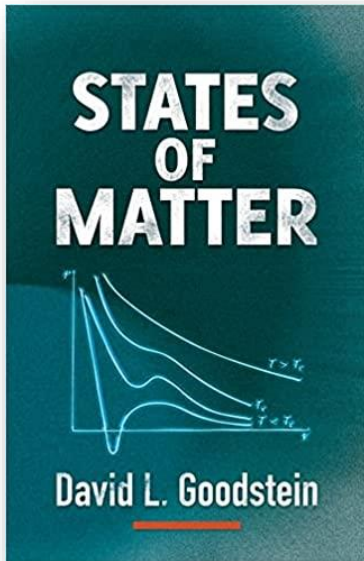
Pressure Dependence of G

$$\Delta G_m = \int_{p_i}^{p_f} V_m dp$$

1. Condensed phase: $V_m \approx \text{constant}$
2. Gas phase: $V_m = V/n = RT/p$

6. Statistical Mechanics





1.1 INTRODUCTION: THERMODYNAMICS AND STATISTICAL MECHANICS OF THE PERFECT GAS

Ludwig Boltzmann, who spent much of his life studying statistical mechanics, died in 1906, by his own hand. Paul Ehrenfest, carrying on the work, died similarly in 1933. Now it is our turn to study statistical mechanics.

Goodstein, *States of Matter* (1985)

Chapter Overview

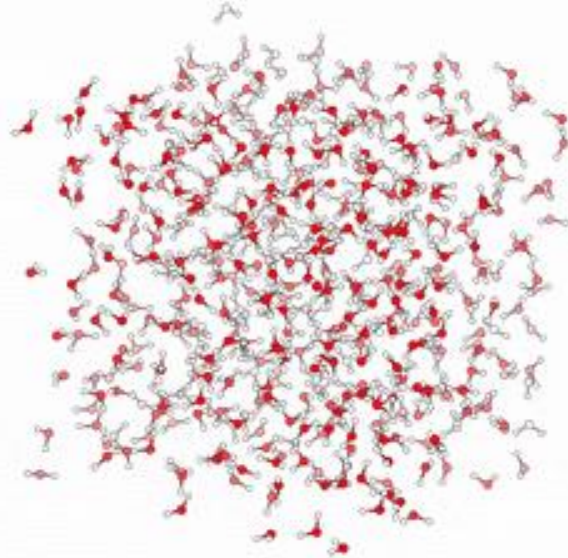
- Maxwell-Boltzmann statistics
 - Boltzmann factor
 - Partition function and thermodynamic quantities
- Classical statistical mechanics
 - Phase space
 - Calculation of partition function
- Applications
 - Perfect gas and real gas
 - Equipartition theorem



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Statistical Mechanics

Thermodynamics ($p, V, T, \dots$)



“Statistics” of
individual particles

(Image: https://commons.wikimedia.org/wiki/File:MD_water.gif)

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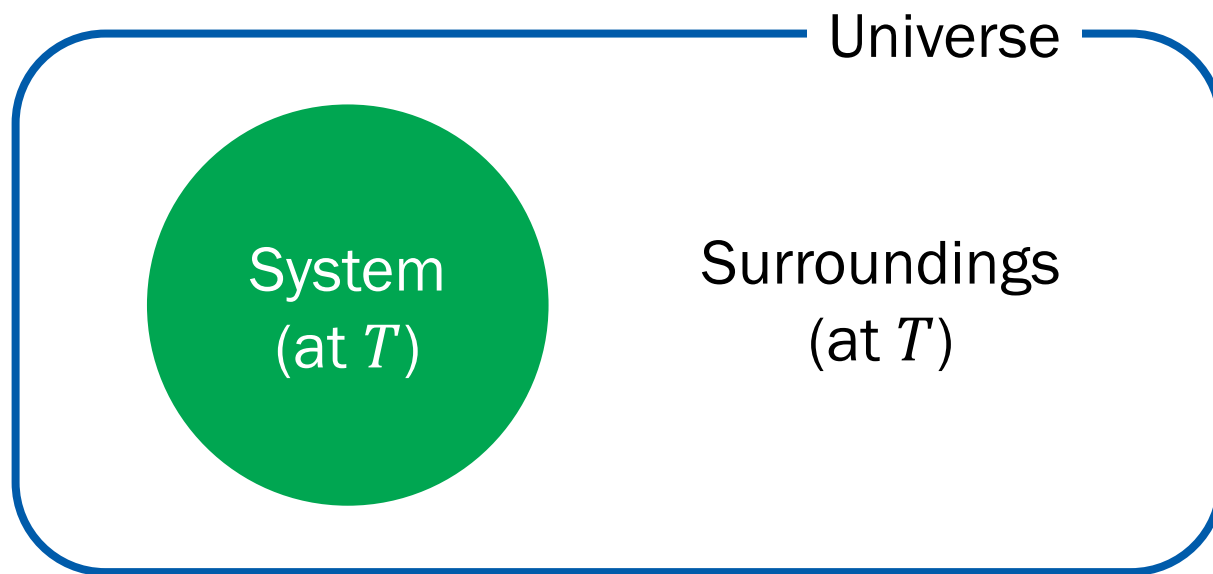
Maxwell-Boltzmann Statistics

Average distribution of independent particles
of a system in **thermal equilibrium**

Described by the **Boltzmann factor**

System Microstates

For a closed system with constant V and n ,



Consider two microstates s_1 and s_2 .

Probabilities

- Probability of having s_i is proportional to the number of microstates in the surroundings

$$P(s_i) \propto W_{\text{surr}}(s_i)$$

- Ratio between probabilities of having s_1 and s_2

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\text{surr}}(s_2)}{W_{\text{surr}}(s_1)}$$

Entropy Comes in

$$\frac{P(s_2)}{P(s_1)} = \frac{W_{\text{surr}}(s_2)}{W_{\text{surr}}(s_1)}$$

Using the statistical definition of entropy, $S = k_B \ln W$,

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{S_{\text{surr}}(s_2)/k_B}}{e^{S_{\text{surr}}(s_1)/k_B}} = e^{\{S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)\}/k_B}$$

What is $S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)$?

Entropy of Surroundings

Recall the fundamental equation of thermodynamics:

$$dU = TdS - pdV = TdS \text{ (at constant } V\text{)}$$

For constant T , $\Delta U = T\Delta S$ and

$$\begin{aligned} S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1) &= \frac{1}{T} \{U_{\text{surr}}(s_2) - U_{\text{surr}}(s_1)\} \\ &= -\frac{1}{T} \{U(s_2) - U(s_1)\} \end{aligned}$$

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Written in System Variables

$$\frac{P(s_2)}{P(s_1)} = e^{\{S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1)\}/k_B}$$

$$S_{\text{surr}}(s_2) - S_{\text{surr}}(s_1) = -\frac{1}{T}\{U(s_2) - U(s_1)\}$$

Combining the two equations,

$$\frac{P(s_2)}{P(s_1)} = e^{-\{U(s_2) - U(s_1)\}/k_B T} = \frac{e^{-U(s_2)/k_B T}}{e^{-U(s_1)/k_B T}}$$

Boltzmann Factor

$$\frac{P(s_2)}{P(s_1)} = \frac{e^{-U(s_2)/k_B T}}{e^{-U(s_1)/k_B T}}$$

In general, for a microstate s with energy E ,

$$P(s) \propto e^{-E(s)/k_B T} \text{ (Boltzmann factor)}$$

or, using a proportionality constant z ,

$$P(s) = \frac{1}{z} e^{-E(s)/k_B T}$$

Determination of z

$$P(s) = \frac{1}{z} e^{-E(s)/k_B T}$$

Using the normalization condition, $\sum_s P(s) = 1$,

$$1 = \sum_s \frac{1}{z} e^{-E(s)/k_B T} = \frac{1}{z} \sum_s e^{-E(s)/k_B T}$$

$$z = \sum_s e^{-E(s)/k_B T}$$

(Partition function)

Brief Illustration 13A.2

Suppose that two conformations of a molecule differ in energy by 5.0 kJ mol⁻¹ (corresponding to 8.3×10^{-21} J for a single molecule), so conformation A lies at energy 0 and conformation B lies at $\varepsilon = 8.3 \times 10^{-21}$ J. What is the proportion of molecules in conformation B at 293 K?

$$E(A)/k_B T = 0 / \{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (293 \text{ K})\} = 0$$

$$E(B)/k_B T = (8.3 \times 10^{-21} \text{ J}) / \{(1.381 \times 10^{-23} \text{ J K}^{-1}) \times (293 \text{ K})\} = 2.05$$

The partition function is

$$z = \sum_s e^{-E(s)/k_B T} = e^{-E(A)/k_B T} + e^{-E(B)/k_B T} = e^0 + e^{-2.05} = 1.13$$

Brief Illustration 13A.2

$$z = 1.13$$

Hence, the proportion of molecules in conformation B is

$$P(\text{B}) = \frac{1}{z} e^{-E(\text{B})/k_{\text{B}}T} = \frac{1}{1.13} e^{-2.05} = 0.11$$

Inverse Temperature

$$\beta = \frac{1}{k_{\text{B}}T}$$

$$Z = \sum_i e^{-E_i/k_{\text{B}}T} = \sum_i e^{-\beta E_i}$$

Probability Distributions

	Discrete	Continuous
Variable	$k = 1, 2, \dots, N$	x : real in $[x_1, x_2]$
Probability	$p(k)$	$p(x)dx$ probability density ↘
Normalization	$\sum_{k=1}^N p(k) = 1$	$\int_{x_1}^{x_2} p(x) dx = 1$
Average	$\langle k \rangle = \sum_{k=1}^N kp(k)$	$\langle x \rangle = \int_{x_1}^{x_2} xp(x) dx$
Expectation	$\langle f \rangle = \sum_{k=1}^N f(k)p(k)$	$\langle f \rangle = \int_{x_1}^{x_2} f(x)p(x) dx$

Average Energy

According to the probability theory, the average value of k is

$$\langle k \rangle = \sum_i p_i k_i$$

- Average energy (= internal energy)

$$\langle E \rangle = \sum_i P(E_i) E_i = \sum_i \frac{E_i e^{-\beta E_i}}{Z}$$

Average Energy

$$\langle E \rangle = \sum_i \frac{E_i e^{-\beta E_i}}{z}$$

However,

$$\frac{\partial \ln z}{\partial \beta} = \frac{1}{z} \frac{\partial z}{\partial \beta} = -\frac{1}{z} \sum_i E_i e^{-\beta E_i}$$

Hence,

$$\langle E \rangle = U = -\frac{\partial \ln z}{\partial \beta}$$

Entropy

$$S = k_B \ln W$$

Assumption: all microstates are equally probable!

What if not ...?

Shannon Entropy

Consider a system with the following properties:

- t microstates
- each microstate i with probability p_i
- N independent replicates ($N \rightarrow \infty$)

Among N , the number of finding microstate i is $n_i = Np_i$

Then, the number of combinations is

$$W = \frac{N!}{n_1! n_2! \cdots n_t!}$$

Shannon Entropy

$$W = \frac{N!}{n_1! n_2! \cdots n_t!}$$

$$S = k_B \ln W = k_B \ln \frac{N!}{n_1! n_2! \cdots n_t!}$$

By Stirling's approximation,

$$S \approx k_B (N \ln N - \sum_i n_i \ln n_i) = -k_B \sum_i n_i \ln(n_i/N)$$

$$S = -Nk_B \sum_i p_i \ln p_i$$

Hence, entropy per system is $S = -k_B \sum_i p_i \ln p_i$

Entropy

$$S = -k_B \sum_i p_i \ln p_i \text{ (Shannon entropy)}$$

Substitution of $p_i = P(E_i) = \frac{1}{z} e^{-\beta E_i}$ leads to

$$S = -k_B \sum_i \frac{e^{-\beta E_i}}{z} (-\beta E_i - \ln z)$$

$$S = \beta k_B \sum_i \frac{E_i e^{-\beta E_i}}{z} + \frac{k_B \ln z}{z} \sum_i e^{-\beta E_i}$$

$$S = \frac{\langle E \rangle}{T} + k_B \ln z$$

Helmholtz Energy (?!)

$$S = \frac{\langle E \rangle}{T} + k_B \ln z$$

Therefore,

$$-k_B T \ln z = \langle E \rangle - TS = A$$

Partition Function and ...

- Helmholtz energy: $A = -k_{\text{B}}T \ln z$
- Internal energy: $U = -\partial(\ln z)/\partial\beta$
- Entropy: $S = (U - A)/T$

“A partition function contains
all the **thermodynamic information** about a system.”

Derived Functions

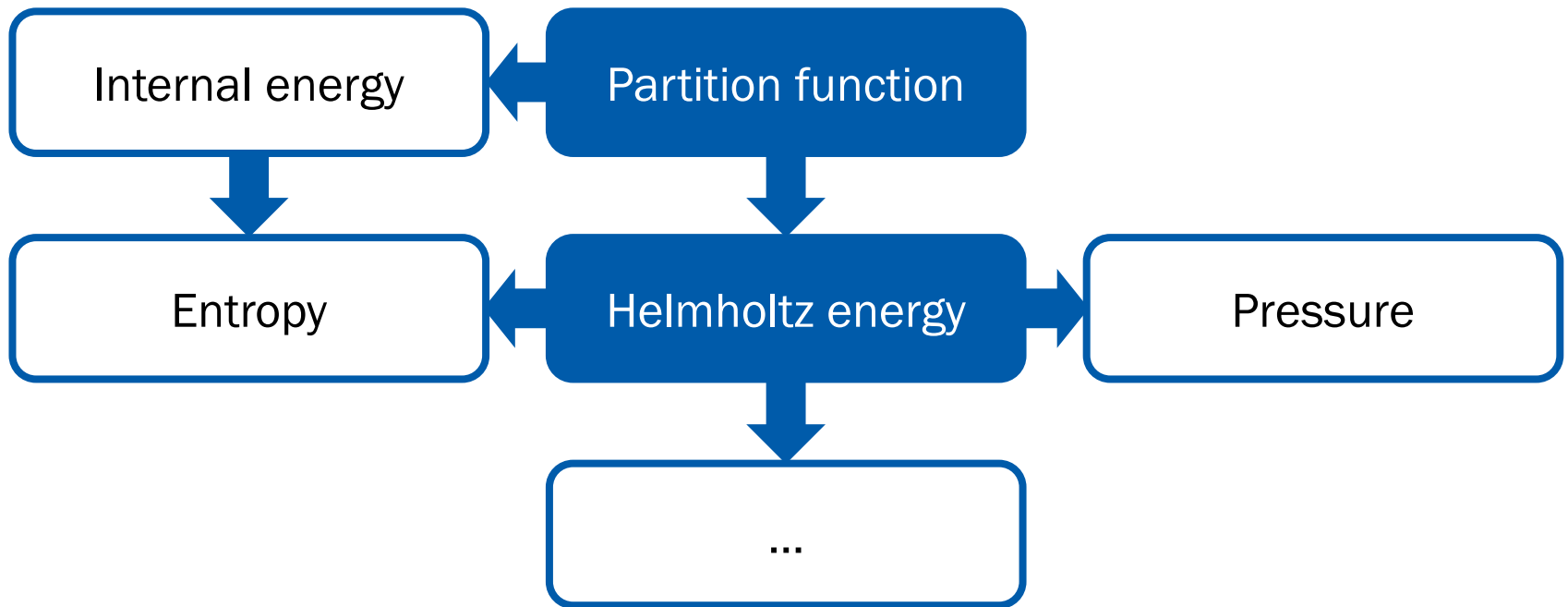
We know Helmholtz energy from the partition function,

$$A = -k_{\text{B}}T \ln z$$

Using $dA = -SdT - pdV$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_V, \quad p = -\left(\frac{\partial A}{\partial V}\right)_T$$

z and Thermodynamics



Example: Spin System

Assume that we have a single spin system: up ($\uparrow$) or down ($\downarrow$)

$$E(\uparrow) = +\frac{\epsilon}{2}, \quad E(\downarrow) = -\frac{\epsilon}{2}$$

The partition function is

$$z = \sum_s e^{-E(s)/k_B T} = e^{-E(\uparrow)/k_B T} + e^{-E(\downarrow)/k_B T}$$

$$z = e^{-\epsilon/2k_B T} + e^{+\epsilon/2k_B T}$$

Example: Spin System

$$z = e^{-\epsilon/2k_B T} + e^{+\epsilon/2k_B T}$$

- Internal energy $U = -\partial(\ln z)/\partial\beta$

$$U = -\frac{1}{z} \frac{\partial z}{\partial \beta} = -\frac{1}{z} \frac{\partial}{\partial \beta} (e^{-\beta\epsilon/2} + e^{\beta\epsilon/2})$$

$$U = -\frac{1}{z} \left(-\frac{\epsilon}{2} e^{-\beta\epsilon/2} + \frac{\epsilon}{2} e^{\beta\epsilon/2} \right)$$

$$U = -\frac{\epsilon}{2} \frac{e^{\beta\epsilon/2} - e^{-\beta\epsilon/2}}{e^{\beta\epsilon/2} + e^{-\beta\epsilon/2}} = -\frac{\epsilon}{2} \frac{e^{\epsilon/2k_B T} - e^{-\epsilon/2k_B T}}{e^{\epsilon/2k_B T} + e^{-\epsilon/2k_B T}}$$

Example: Spin System

$$z = e^{-\epsilon/2k_B T} + e^{+\epsilon/2k_B T}$$

- Helmholtz energy $A = -k_B T \ln z$

$$A = -k_B T \ln(e^{-\epsilon/2k_B T} + e^{\epsilon/2k_B T})$$

- Entropy $S = (U - A)/T = -(\partial A/\partial T)_V$

$$S = -\frac{\epsilon}{2T} \frac{e^{\epsilon/2k_B T} - e^{-\epsilon/2k_B T}}{e^{\epsilon/2k_B T} + e^{-\epsilon/2k_B T}} + k_B \ln(e^{-\epsilon/2k_B T} + e^{\epsilon/2k_B T})$$

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Classical Statistical Mechanics

What is a **microstate** of a classical particle?



Position and Velocity

Each particle with mass m has its own

- Position $\vec{r} = (x, y, z)$
- Velocity $\vec{v} = (v_x, v_y, v_z)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right)$$

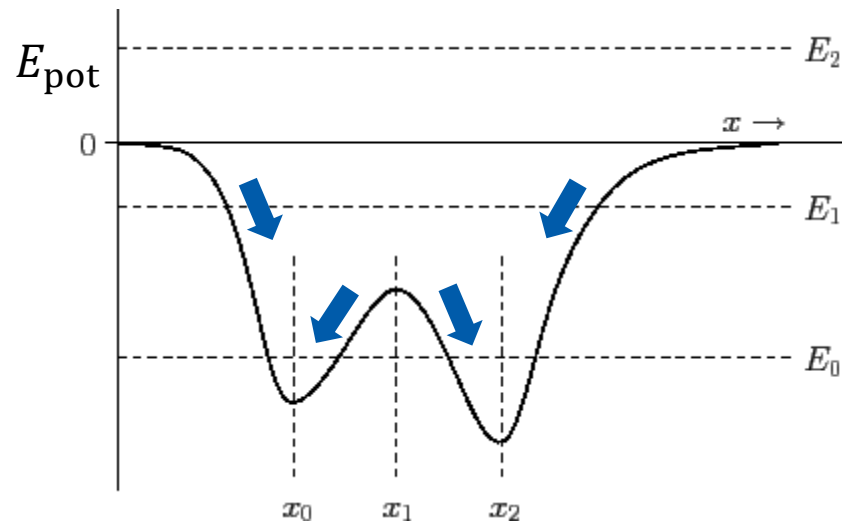
With the velocity, we can define:

- Speed $v = |\vec{v}| = (v_x^2 + v_y^2 + v_z^2)^{1/2}$
- Kinetic energy $E_k = mv^2/2 = m(v_x^2 + v_y^2 + v_z^2)/2$

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Force $\vec{F}$: negative slope of potential energy ($-\nabla E_{\text{pot}}$)



(Image: <http://farside.ph.utexas.edu/teaching/336k/Newtonhtml/node16.html>)

Newton's Second Law

$$\vec{F} = m\vec{a}$$

- Acceleration $\vec{a}$: second time-derivative of position

$$\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$$

$$-\nabla E_{\text{pot}} = m \frac{d^2\vec{r}}{dt^2}$$

↑
given

Second-Order ODE

$$\frac{d^2 \vec{r}}{dt^2} = - \frac{\nabla E_{\text{pot}}}{m}$$

Two conditions are required to obtain the full dynamics:

(initial) position and (initial) velocity (= momentum)

Phase Space

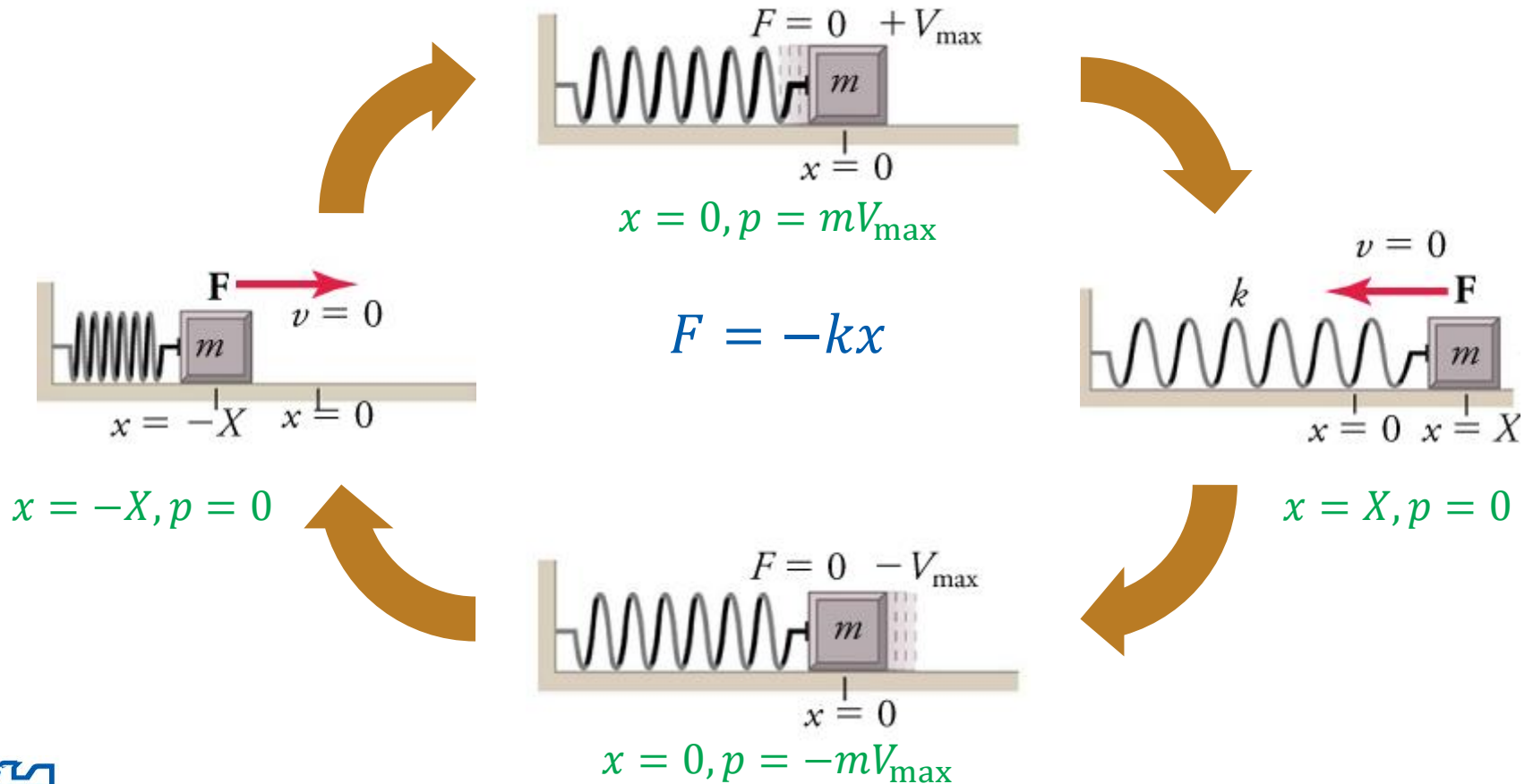
Space of **all possible microstates** of a system

- Classical mechanics:

Space of all possible values of
position and **momentum** variables

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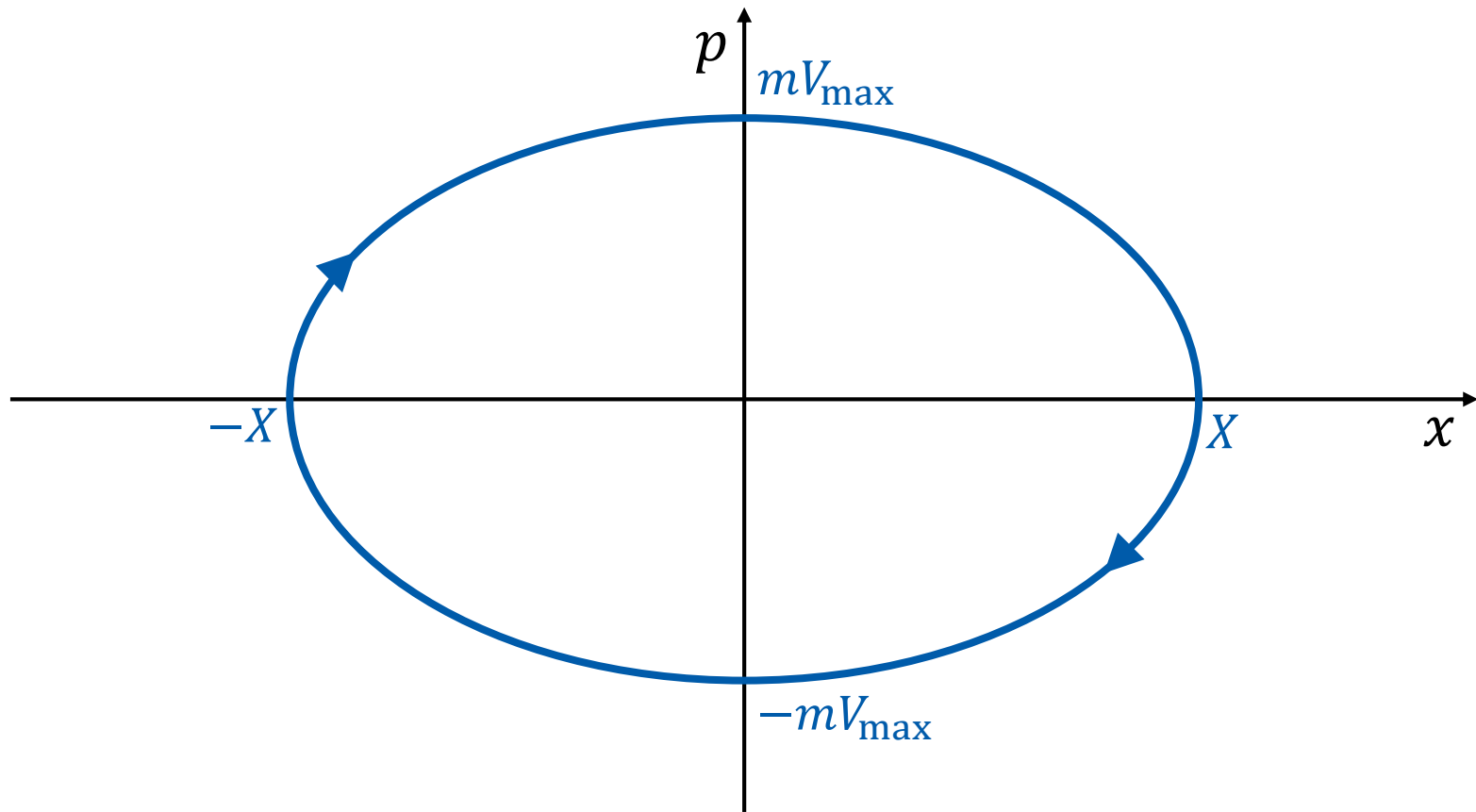
1-Dimensional Spring



(Image: <https://openstax.org/books/physics/pages/5-5-simple-harmonic-motion>)

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Phase Space Representation



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Microstate of Classical Particle

